Factors that Influence Diffusion

Exam code: 0610 & 0970

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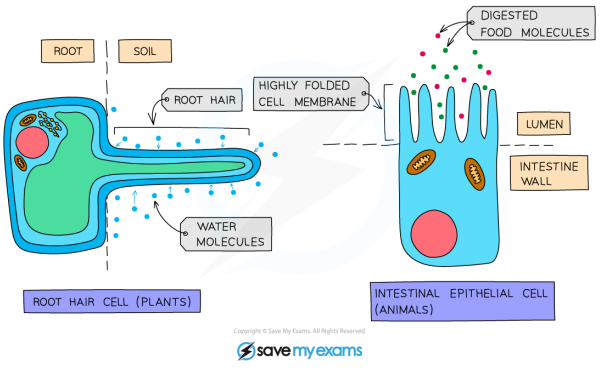
Factors that Influence Diffusion

Surface area to volume ratio

- The **bigger** a cell or structure is, the **smaller its surface area to volume ratio** is, slowing down the rate at which substances can move across its surface
- Many cells which are adapted for diffusion have **increased surface area** in some way - eg root hair cells in plants (which absorb water and mineral ions) and cells lining the ileum in animals (which absorb the products of digestion)



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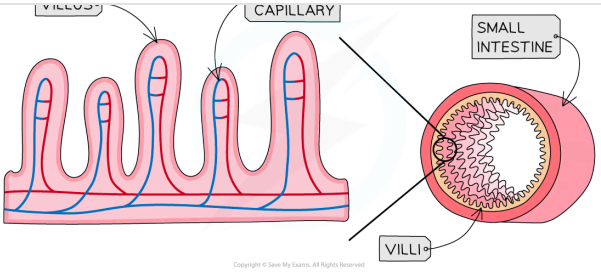

The highly folded surface of the small intestine increases its surface area

Distance

- The smaller the distance molecules have to travel the faster transport will occur
- This is why blood capillaries and alveoli have walls which are only one cell thick, ensure the rate of diffusion across them is as fast as possible

Temperature

- The higher the temperature, the faster molecules move as they have more energy
- This results in more collisions against the cell membrane and therefore a faster rate of movement across them

Concentration Gradient

- The greater the difference in concentration on either side of the membrane, the faster movement across it will occur
- This is because on the side with the higher concentration, more random collisions against the membrane will occur

Examiner Tips and Tricks

You should have carried out investigations into the factors that influence the rate of diffusion and as so should be able to use the information above to **explain experimental results** in an exam. You should also be able to **plan and carry out an experiment** which can investigate the effect of one of these factors.

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